What if space doesn’t really exist?

That’s the question behind this little paper. Not a provocation — a serious hypothesis, in the spirit of Van Raamsdonk, Verlinde, and a few others who enjoy dismantling the foundations of physics.
The central idea is simple:
the geometry of spacetime might not be fundamental.
It could emerge from correlations between quantum systems.
Two particles that share a lot of information would be “close”.
Two particles with no shared information would be “far”.
Space would be a map of relationships, not a pre-existing container.
Each particle would carry two kinds of degrees of freedom:
visible ones (the part we can measure),
private ones (internal dimensions, inaccessible directly).
Decoherence — the mechanism that destroys quantum superpositions — would play a double role here: it destroys coherence, yes, but it also selects the geometry that survives.
From this, three heuristic predictions follow:

Dark matter would be the geometric effect of invisible private correlations.
Dark energy would be the mathematical consequence of the progressive loss of visible correlations — the universe expands because it “forgets”.

Black holes would be regions where the density of information exceeds what a 3D geometry can represent.

A small numerical model on 4 qubits reproduces the three predicted regimes:
stable geometry, transition, and collapse.

It’s speculative. It’s honest about its limits. And it holds together.

— L. Guizelin, 2026

Emergent quantum geometry

Space as a relational property

Each particle
Visible → builds geometry
Private → stores information · [H = H_vis ⊗ H_priv]

Distance = correlation
Highly correlated → close
Weakly correlated → far · [D_ij = exp(−α · I_ij)]

Decoherence
Destroys superpositions · Selects surviving geometry

Three regimes according to rate γ

γ low
Stable geometry
o—o—o—o
D_eff high

γ medium
Transition
o—o o—o
D_eff falling

γ high
Collapse
o o o o
D_eff → 1

Cosmological implications

Dark matter
Private correlations
→ hidden geometry

Dark energy
Loss of information
→ D_ij grows → universe expands

Black holes
Critical info density
→ 3D fabric breaks down

Mathematical pipeline

State ρ(t) → Entropy S → Info I_ij → D_eff → Geometry x_i(t)

In the tradition of: Zurek (decoherence) · Van Raamsdonk (entanglement ↔ space)
Verlinde (entropic gravity) · Penrose/Diósi (objective reduction)
Approach: bottom-up · operational · simulable · falsifiable

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